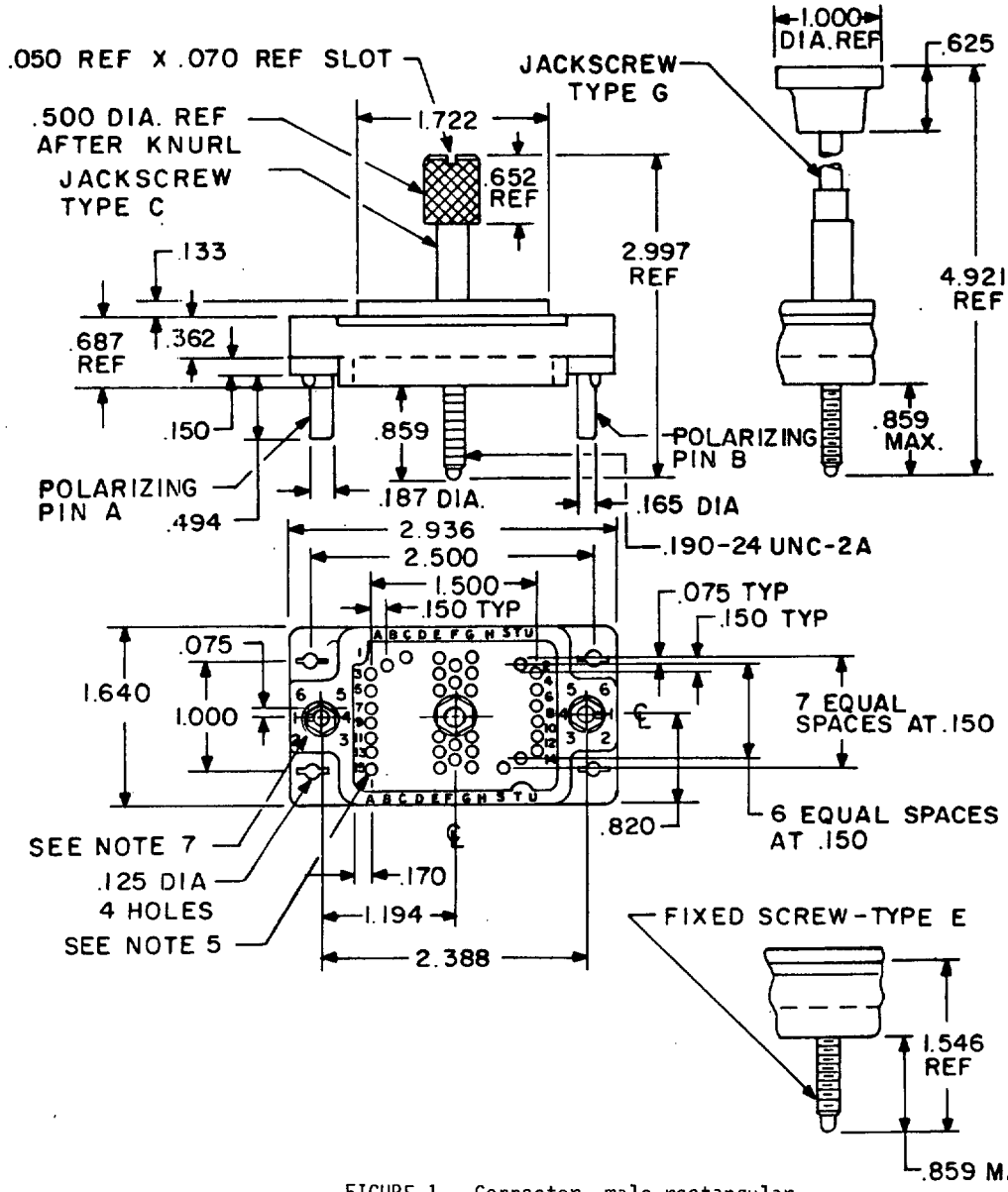
MILITARY SPECIFICATION SHEET

# CONNECTOR, ELECTRICAL, INSERT (INSULATOR), MALE RECTANGULAR, POLARIZED, CENTER SCREW LOCK, FOR 75 REMOVABLE CONTACTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-28731.



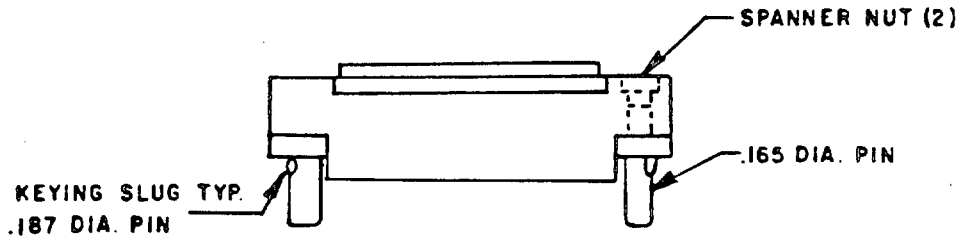

| INCHES | MM |
|---|---|
| .050 | 1.27 |
| .070 | 1.78 |
| .075 | 1.90 |
| .125 | 3.18 |
| .133 | 3.38 |
| .150 | 3.81 |
| .165 | 4.19 |
| .170 | 4.32 |
| .187 | 4.75 |
| .190 | 4.83 |
| .362 | 9.19 |
| .494 | 12.55 |
| .500 | 12.70 |
| .625 | 15.88 |
| .652 | 16.56 |
| .687 | 17.45 |
| .820 | 20.83 |
| .859 | 21.82 |
| 1.000 | 25.40 |
| 1.194 | 30.33 |
| 1.500 | 38.10 |
| 1.546 | 39.27 |
| 1.640 | 41.66 |
| 1.722 | 43.74 |
| 2.388 | 60.66 |
| 2.500 | 63.50 |
| 2.936 | 74.57 |
| 2.997 | 76.12 |
| 4.921 | 125.00 |

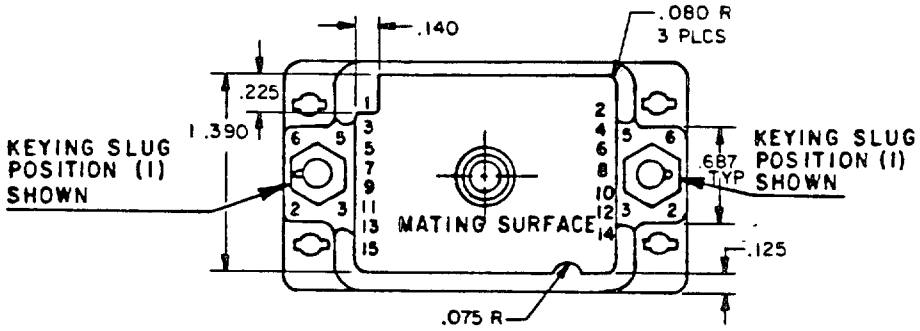
FIGURE 1. Connector, male rectangular.

(E) denotes changes



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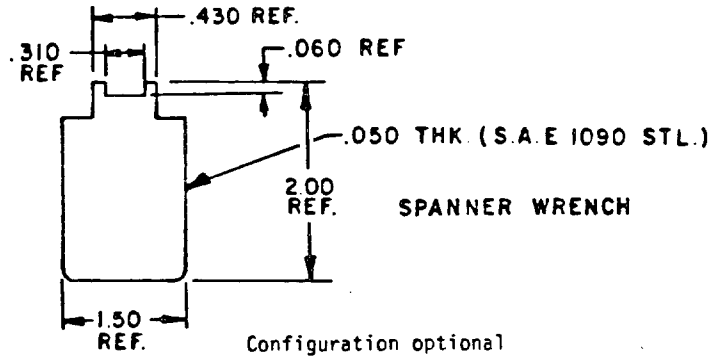




POLARIZING POSITION CHANGE

Keying position on both polarizing pins can be rotated to desired position by loosening spanner nut (counter-clockwise), resetting pin in one of six positions then retightening spanner nut (clockwise).

NOTE: Overtightening of spanner nut can fracture connector. Spanner keying wrench shown below.

| INCHES | MM |
|---|---|
| .050 | 1.27 |
| .060 | 1.52 |
| .075 | 1.90 |
| .080 | 2.03 |
| .125 | 3.17 |
| .140 | 3.56 |
| .225 | 5.71 |
| .310 | 7.87 |
| .430 | 10.92 |
| .687 | 17.45 |
| 1.390 | 35.31 |
| 1.500 | 38.10 |
| 2.000 | 50.80 |

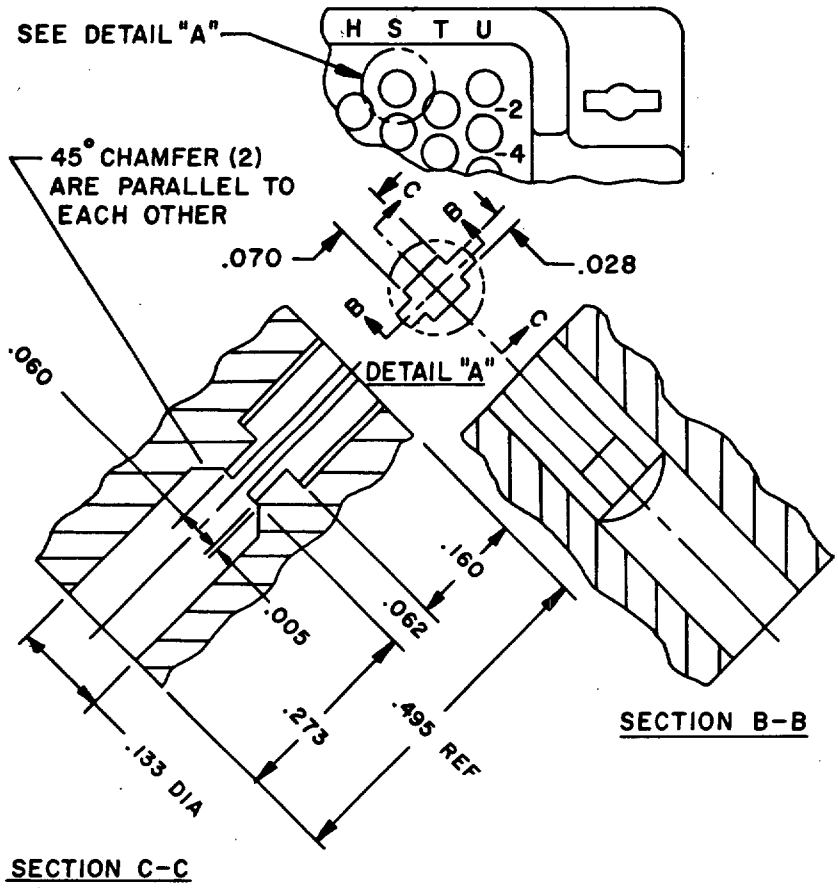



Configuration A - Connector polarization guide (layout)

FIGURE 1. Connector, male rectangular - Continued.



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| INCHES | MM |
|---|---|
| .005 | 0.13 |
| .028 | 0.71 |
| .060 | 1.52 |
| .062 | 1.57 |
| .070 | 1.78 |
| .133 | 3.38 |
| .160 | 4.06 |
| .273 | 6.93 |
| .495 | 12.57 |

Configuration B - Contact hole detail

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerances between any two contact centers shall be ±.004 (0.10 mm).
4. Flatness, squareness, and parallelism shall be within dimensional tolerances.
5. Contact hole details shall be in accordance with configuration B.
6. Mating connector information:
   M28731/1-0109, -0110, -0112 and -0113 mate with M28731/2-0073, -0075 and -0077.
   M28731/1-0111, -0114 and -0115 mate with M28731/2-0074 and 0076.
(E) 7. Polarization information:
   Polarization pins are provided for indexing to preclude intermixing of adjacent connectors. By using a standard type spanner wrench, polarization pins can be rotated to desired position. For parts supplied directly to the government, polarization will be in the 1-1 position. For other polarization positions see connector polarization guide configuration A (see marking, note 13d). For direct government acquisition a tool is supplied in the unit package for polarization pin adjustment.

FIGURE 1. Connector, male rectangular - Continued.



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8. Contact and terminal designations:
Contacts and terminals shall be designated on the insulator using letters in the location shown on figure 1.
9. Tolerance shall be ±.010 (0.25 mm) unless otherwise specified.
10. Patent notice:
Patent No. 3,248,686 owned by Elco Corporation. The government has a royalty-free license under the above patent for the benefit of manufacturers of items called for in this specification, either for the government or for use in equipment to be delivered to the government.
11. For direct government acquisition crimp contacts M28731/20-0001 are furnished in a plastic bag in each unit package (see marking, note 13a).
12. For direct government acquisition solder hole and wrappost contacts M28731/25-0001 and 26-0001 are factory installed in connector body (see marking, note 13a).
13. Marking:
    a. Connectors supplied directly to the government shall be marked with the military part number (see table I). Example: M28731/1-0110.
    b. Connectors supplied with contacts to government end users shall be marked with the military part number (see table I). Example: M28731/1-0110.
    Ⓔ c. Connectors supplied without contacts to government end users shall be marked as follows: Example: M28731/1G (screw type from table I). The user shall complete the remaining portion of the military part number as shown in table I based upon contact type he assembles. Example: M28731/1G-0110.
    d. Polarization position marking is not required unless otherwise specified. When required polarization marking shall be in accordance with the connector polarization guide (see configuration A). The marking shall appear following the part number.
14. Applicable contact insertion and removal tools:

| | |
|---|---|
| Insertion | MIL-I-81969/13-01 |
| Removal | MIL-I-81969/11-01 |

15. When double wire crimp contact MIL-C-28731/23 is required for use with the connector, it shall be ordered separately.
16. Rectangular shields shall conform to MIL-C-28731/7-0001 or -0002 and must be ordered separately.

FIGURE 1. Connector, male rectangular - Continued.



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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Mating and unmating forces (torque): 100 inch-ounces, maximum.

Mechanical shock (specified pulse): Applicable.

Part number: See table I.

TABLE I. Part number information.

| Military part number | Actuating hardware | Polarizing pin position A .187 Dia. | Polarizing pin position B .165 Dia. | Contact type |
|---|---|---|---|---|
| M28731/1-0109 | Jackscrew Type C | 1 | 1 | Crimp MIL-C-28731/20 1/ |
| M28731/1-0110 | Jackscrew Type G | 1 | 1 | Crimp MIL-C-28731/20 1/ |
| M28731/1-0111 | Fixed screw Type E | 1 | 1 | Crimp MIL-C-28731/20 1/ |
| M28731/1-0112 | Jackscrew Type C | 1 | 1 | Solder hole MIL-C-28731/25 2/ |
| M28731/1-0113 | Jackscrew Type G | 1 | 1 | Solder hole MIL-C-28731/25 2/ |
| M28731/1-0114 | Fixed screw Type E | 1 | 1 | Solder hole MIL-C-28731/25 2/ |
| M28731/1-0115 | Fixed screw Type E | 1 | 1 | Wrappost MIL-C-28731/26 2/ |

1/ See figure 1, note 11
2/ See figure 1, note 12

NOTE:
M28731/1-0109 supersedes M28731/1-0001 thru M28731/1-0036.
M28731/1-0110 supersedes M28731/1-0037 thru M28731/1-0072.
M28731/1-0111 supersedes M28731/1-0073 thru M28731/1-0108.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, MI
Navy - AS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AL, AV, ME
Navy - MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5935-3488-1)



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